

Simple Practice Quiz

Platform Based Development (PBDE401/PBDV301)

March 22, 2026

Q1. In a Flask application, what happens if `db.session.commit()` is not executed after `db.session.add()`?

- A. The object is automatically stored in the database
- B. The object remains only in memory and is not persisted
- C. SQLAlchemy throws an immediate error
- D. The database connection closes
- E. The object becomes read-only

Q2. In relational databases, a table used to implement a many-to-many relationship is called:

- A. Reference table
- B. Junction table
- C. Parent table
- D. Session table
- E. Cascade table

Q3. Which Flask feature prevents Cross-Site Request Forgery attacks when using forms?

- A. Flask-Login
- B. Flask-SQLAlchemy
- C. CSRF token in Flask-WTF
- D. Jinja Templates
- E. SQLite

Q4. In the route `'/product/<int:id>'`, what does `<int:id>` indicate?

- A. The id parameter must be an integer
- B. The id is optional

- C. The id references a database column
- D. The id must be a string
- E. The id is a session variable

Q6. Which SQLAlchemy command retrieves all rows from a table?

- A. Model.select()
- B. Model.query.all()
- C. Model.fetch()
- D. Model.records()
- E. Model.getAll()

Q7. Which concept refers to adding more servers to distribute system load?

- A. Vertical scaling
- B. Horizontal scaling
- C. Server caching
- D. Thread pooling
- E. Virtualization

Q8. Vertical scaling refers to:

- A. Adding more servers
- B. Increasing hardware resources of a single server
- C. Increasing bandwidth
- D. Using CDN nodes
- E. Using microservices

Q9. Which web constraint refers to websites failing to adapt to different screen sizes?

- A. Internet reliance
- B. Browser incompatibility
- C. Lack of responsiveness
- D. Reduced performance

E. Data compliance

Q10. Which solution best improves browser compatibility?

A. Removing JavaScript

B. Using cross-browser frameworks and testing

C. Increasing server RAM

D. Removing CSS

E. Using static HTML only

Q11. A university deploys a Flask-based learning platform hosted on a single cloud server. During exam registration periods, thousands of students attempt to submit forms simultaneously, causing long response times and server crashes. Which architectural change would MOST effectively improve the platform's scalability and reliability?

A. Increase the HTML rendering speed

B. Upgrade the single server to a larger machine (vertical scaling)

C. Deploy multiple application instances behind a load balancer (horizontal scaling)

D. Disable database logging

E. Move all data processing to the client browser

Q12. A Flask web application uses SQLite for storing customer orders. After the platform becomes popular, developers observe frequent database locking errors when many users place orders simultaneously. What is the most appropriate solution?

A. Replace SQLite with PostgreSQL or MySQL

B. Remove database relationships

C. Store orders temporarily in browser cookies

D. Disable transaction management

E. Convert the application into a static website

Q13. A company launches a SaaS CRM platform where multiple businesses use the same system but cannot access each other's data. All customers share the same application infrastructure and database schema. Which SaaS architecture model is being implemented?

A. Single-tenant SaaS

B. Multi-tenant SaaS with logical isolation

- C. Peer-to-peer SaaS
- D. Distributed SaaS
- E. Local SaaS deployment

Q14. A mobile food delivery app continuously uses GPS tracking, background network communication, and push notifications. Users complain that the application significantly drains their phone batteries. Which mobile operating system component is primarily responsible for managing this issue?

- A. Device driver management
- B. Power and battery management
- C. Application UI management
- D. Process virtualization
- E. Database optimization

Q15. A web platform is deployed globally, but users in remote regions experience slow page load times, even though the main server is powerful. Which solution would most effectively reduce latency for global users?

- A. Increase HTML compression
- B. Deploy a Content Delivery Network (CDN)
- C. Reduce database size
- D. Remove CSS animations
- E. Increase server RAM

Q16. A developer builds a Flask blogging platform where deleting a user must automatically remove all blog posts written by that user. Which SQLAlchemy configuration best supports this requirement?

- A. Remove foreign keys
- B. Use cascade delete relationships
- C. Store posts in JSON files
- D. Disable database constraints
- E. Delete posts manually in JavaScript

Q17. A government web portal must comply with strict data protection laws requiring user consent before collecting personal information. Which system feature would best support legal compliance?

- A. Disable user accounts
- B. Implement explicit consent forms and privacy policies

- C. Store user passwords in plain text
- D. Allow unrestricted data access to administrators
- E. Remove encryption

Q18. A web application is accessed from smartphones, tablets, and desktops. Users report that the interface appears distorted on smaller screens. Which design approach best solves this problem?

- A. Fixed-width layouts
- B. Responsive web design using CSS media queries
- C. Increasing server bandwidth
- D. Removing JavaScript
- E. Increasing image resolution

Q19. A large streaming platform experiences heavy traffic spikes during live events. Engineers need a system that automatically distributes incoming traffic across servers. Which infrastructure component should they implement?

- A. Database indexing
- B. Load balancer
- C. HTML templating engine
- D. CSS preprocessing
- E. Local browser caching

Q20. A mobile banking application must prevent other installed apps from accessing sensitive user data stored by the application. Which mobile OS security mechanism provides this protection?

- A. Bluetooth encryption
- B. Application sandboxing
- C. GPU isolation
- D. File compression
- E. Background synchronization

Q21. A developer notices that a Flask search feature becomes extremely slow as the database grows. The query uses:

SELECT * FROM products WHERE name LIKE '%phone%'

Which database optimization would most likely improve performance?

- A. Remove pagination
- B. Add indexing to the searchable column
- C. Move the query to JavaScript
- D. Disable database constraints
- E. Increase HTML caching

Q22. A startup launches a global SaaS platform that stores sensitive healthcare data. The company must ensure data cannot be accessed by unauthorized tenants. Which design strategy best ensures data isolation?

- A. Store all tenant data in one shared table without identifiers
- B. Use tenant identifiers and strict access control
- C. Disable authentication
- D. Use browser cookies for data storage
- E. Store data on client devices

Q23. A mobile app built using native Android development must also run on iOS. The company wants to reuse as much code as possible without losing performance. Which development approach best meets this requirement?

- A. Native development for each platform
- B. Cross-platform frameworks such as Flutter or React Native
- C. Assembly programming
- D. Kernel-level development
- E. Static web pages

Q24. A Flask application depends heavily on server-side processing. When the internet connection drops, users cannot access the application at all. Which solution best reduces the internet reliance constraint?

- A. Disable authentication
- B. Convert the application into a Progressive Web App (PWA) with offline caching
- C. Increase server storage

- D. Remove CSS files
- E. Convert the application into a desktop executable

Q25. A rapidly growing web platform experiences performance issues because the single application server cannot handle increasing traffic. Which scaling strategy would provide the best long-term scalability?

- A. Increase server RAM repeatedly (vertical scaling only)
- B. Add more application servers and distribute traffic (horizontal scaling)
- C. Disable database queries
- D. Store user data in HTML files
- E. Move all logic to JavaScript